Audit: book examples against the Mojo port

The book prints 77 output boxes ("Output from C++ program" and "Output from Matlab program"). `audit/book_examples.mojo` reruns the program of every box with the `finrecipes` package and this table compares each printed number with the port's result. Inputs that Matlab merely echoes back are left out.

The Monte Carlo examples (boxes 47 to 51, 57 and 77) are run with the package's emulation of the GNU C library `rand()`, seeded with 1 at the start of each box, which is the random sequence the book's programs used. They are therefore compared digit for digit, like every other number.

How to read the result column

- **match**: the port agrees with the book to within half a unit of the last printed digit.
- **last digit**: the port differs by at most one unit in the last printed digit.
- **DIFFERS**: the port gives a different number.
- **not comparable**: nothing in the port to compare with (reason given).

Summary

Boxes: 77. Printed numbers compared: 234.

Box result	Boxes
match	75
differs	1
not comparable	1

Number result	Numbers
match	232
DIFFERS	2
not comparable	1

Numbers that do not match

Box	Ch.	Quantity	Book	Port	Difference	Result
73	25	Ho-Lee call, flat term structure	6.4857	7.098447	+6.1e-01	DIFFERS
73	25	Ho-Lee call, Nelson Siegel term structure	6.35411	6.9528190	+6.0e-01	DIFFERS

The 77 boxes

Box	Ch.	Source	Example	Numbers compared	Matching	Result
1	3	Matlab	Present value of cash flows at 10%	4	4	match
2	3	C++	Present value, discrete compounding	1	1	match
3	3	Matlab	Net present value and internal rate of return	5	5	match
4	3	C++	Present value and internal rate of return	2	2	match
5	3	Matlab	Converting between compounding conventions	2	2	match
6	4	Matlab	Bond price	4	4	match
7	4	C++	Bond price	1	1	match
8	4	Matlab	Bond price and yield	2	2	match
9	4	C++	Bond price and yield to maturity	2	2	match
10	4	Matlab	Duration and Macaulay duration	4	4	match
11	4	C++	Duration and Macaulay duration	3	3	match
12	4	Matlab	Duration and convexity	4	4	match
13	4	C++	Modified duration and convexity	5	5	match
14	4	Matlab	Duration-adjusted price change	7	7	match
15	4	C++	Bond with continuous compounding	4	4	match
16	5	C++	Term structure transformations	4	4	match
17	5	C++	Flat term structure class	5	5	match
18	5	Matlab	Discount factors from bond prices	12	12	match
19	5	C++	Linear interpolation of yields	6	6	match
20	5	C++	Interpolated term structure class	5	5	match
21	5	C++	Bond price, duration, convexity with a term structure	3	3	match
22	6	Matlab	Portfolio mean and standard deviation	2	2	match
23	6	Matlab	Minimum variance frontier portfolio	13	13	match
24	6	C++	Mean variance calculations	3	3	match
25	6	C++	Frontier portfolio for a given mean	2	2	match
26	7	C++	Futures price	1	1	match
27	8	C++	One and two period binomial call	2	2	match
28	9	Matlab	Black Scholes call	1	1	match
29	9	C++	Black Scholes call	1	1	match
30	9	C++	Black Scholes partial derivatives	5	5	match

Box	Ch.	Source	Example	Numbers compared	Matching	Result
31	9	C++	Implied volatility	2	2	match
32	10	C++	Warrant price	1	1	match
33	11	C++	European call with dividends	2	2	match
34	11	C++	American call with one dividend (Roll-Geske-Whaley)	1	1	match
35	11	C++	Futures option (Black)	1	1	match
36	11	C++	Currency option	1	1	match
37	11	C++	Perpetual American call	1	1	match
38	12	C++	Binomial European and American options	3	3	match
39	12	Matlab	Binomial American call and put	2	2	match
40	12	C++	Binomial hedge parameters	5	5	match
41	12	C++	Binomial with payouts and dividends	3	3	match
42	12	C++	Futures option, binomial	1	1	match
43	12	C++	Currency option, binomial	1	1	match
44	13	C++	Explicit finite differences	2	2	match
45	13	C++	Implicit finite differences	3	3	match
46	13	Matlab	Implicit finite differences, American put	1	1	match
47	14	C++	European call and put by simulation	4	4	match
48	14	C++	Delta by simulation	4	4	match
49	14	C++	Generic simulation pricing	4	4	match
50	14	C++	Control and antithetic variates	8	8	match
51	14	C++	Binary options by simulation	6	6	match
52	15	C++	Johnson approximation, American put	1	1	match
53	15	C++	Barone-Adesi Whaley approximation, American call	1	1	match
54	16	C++	Bermudan put	1	1	match
55	16	C++	Asian geometric average call	1	1	match
56	16	C++	Lookback call	1	1	match
57	16	C++	Path-dependent options by simulation	7	7	match
58	17	C++	Generic binomial call and put	2	2	match
59	17	C++	Generic binomial binary option	1	1	match
60	18	C++	Trinomial American put	1	1	match
61	19	C++	Merton jump diffusion	1	1	match
62	19	C++	Heston stochastic volatility	1	1	match

Box	Ch.	Source	Example	Numbers compared	Matching	Result
63	20	C++	Bond options	3	3	match
64	21	C++	Credit risk, Merton model	1	1	match
65	22	C++	Nelson Siegel term structure	4	4	match
66	22	C++	Cubic spline term structure	4	4	match
67	22	C++	Cox Ingersoll Ross term structure	4	4	match
68	22	C++	Vasicek term structure	4	4	match
69	23	C++	Rendleman Bartter bond option	1	1	match
70	24	C++	Interest rate tree	6	6	match
71	24	C++	Bond price in an interest rate tree	1	1	match
72	24	C++	Callable bond	2	2	match
73	25	C++	Ho-Lee bond option	2	0	differs
74	26	C++	Vasicek bond option	1	1	match
75	27	C++	Boost date and time demo	0	0	not comparable
76	A	C++	Cumulative normal, univariate and bivariate	2	2	match
77	A	C++	Random numbers	10	10	match

Notes on individual boxes

- **Box 73.** The book's C++ Code 25.4 fills T+1 of the T+2 option values at the last tree level and then reads one element past them, so the printed prices depend on a stray memory value. The port fills the missing payoff. `tests/test_ho_lee.mojo` shows that the printed 6.4857 and 6.35411 are exactly what results when the stray value is 0.5.
- **Box 53.** The port reproduces the printed 5.74339, but the listing passes the cost of carry b where the payout rate is expected, and its Newton iteration does not converge for this example. The published Barone-Adesi Whaley value is 3.52, which `option_price_american_call_approximated_baw_corrected` returns.
- **Box 35.** The book labels the output a call, but the program calls `futures_option_price_put_european_black`. The printed 0.851476 is the put price, and the port's put matches it.
- **Box 75.** Demonstrates the Boost C++ date library and prints the clock time of the run. Nothing to port or compare.
- **Box 12.** The Matlab convexity formula uses t squared and gives 6.6272; C++ Code 4.6 (box 13) uses t(t+1) and gives 8.93248. Both are reproduced; the Matlab value is computed inline in the audit program, because the package ports the C++ routine.
- **Box 5.** The book has no routine for these two conversions; they are evaluated inline in the audit program.
- **Box 18.** The discount factors come from `mat_solve`; the conversion to annual rates is evaluated inline.

Every number

Box	Ch.	Quantity	Book	Port	Difference / note	Result
1	3	d(1) at 10%	1.0000	1.000000	+0.0e+00	match
1	3	d(2) at 10%	0.9091	0.909091	-9.1e-06	match
1	3	d(3) at 10%	0.8264	0.826446	+4.6e-05	match
1	3	NPV	30.165	30.16529	+2.9e-04	match
2	3	Present value, 10% discrete	30.1653	30.165289	-1.1e-05	match
3	3	d(1) at 5%	1.0000	1.000000	+0.0e+00	match
3	3	d(2) at 5%	0.9524	0.952381	-1.9e-05	match
3	3	d(3) at 5%	0.9070	0.907029	+2.9e-05	match
3	3	NPV	9.2971	9.297052	-4.8e-05	match
3	3	IRR	0.100000	0.10000000	+0.0e+00	match
4	3	Present value, 5% discrete	9.29705	9.2970522	+2.2e-06	match
4	3	Internal rate of return	0.1	0.100	+0.0e+00	match
5	3	r = 12 log(1+0.15/12) (inline arithmetic)	0.1491	0.149070	-3.0e-05	match
5	3	r4 = 4 (exp(0.12/4)-1) (inline arithmetic)	0.1218	0.121818	+1.8e-05	match
6	4	d(1) at 9%	0.9174	0.917431	+3.1e-05	match
6	4	d(2) at 9%	0.8417	0.841680	-2.0e-05	match
6	4	d(3) at 9%	0.7722	0.772183	-1.7e-05	match
6	4	B	102.53	102.5313	+1.3e-03	match
7	4	bonds price	102.531	102.53129	+2.9e-04	match
8	4	B	102.53	102.5313	+1.3e-03	match
8	4	y	0.090000	0.09000003	+3.3e-08	match
9	4	Bond price, 9% discrete	102.531	102.53129	+2.9e-04	match
9	4	bond yield to maturity	0.09	0.0900	+3.3e-08	match
10	4	B	102.53	102.5313	+1.3e-03	match
10	4	D	2.7390	2.738954	-4.6e-05	match
10	4	y	0.090000	0.09000003	+3.3e-08	match
10	4	DM	2.7390	2.738954	-4.6e-05	match
11	4	bonds price	102.531	102.53129	+2.9e-04	match
11	4	bond duration	2.73895	2.7389536	+3.6e-06	match
11	4	bond macaulay	2.73895	2.7389536	+3.6e-06	match

Box	Ch.	Quantity	Book	Port	Difference / note	Result
12	4	B	102.53	102.5313	+1.3e-03	match
12	4	D	2.7390	2.738954	-4.6e-05	match
12	4	Cx (Matlab formula, inline arithmetic)	6.6272	6.627156	-4.4e-05	match
12	4	newB	100.000	100.00000	-2.8e-14	match
13	4	bonds price	102.531	102.53129	+2.9e-04	match
13	4	bond duration	2.73895	2.7389536	+3.6e-06	match
13	4	bond duration modified	2.5128	2.512801	+1.4e-06	match
13	4	bond convexity	8.93248	8.9324788	-1.2e-06	match
13	4	new bond price	100	100.00	-2.8e-14	match
14	4	d(1) at 9%	0.9174	0.917431	+3.1e-05	match
14	4	d(2) at 9%	0.8417	0.841680	-2.0e-05	match
14	4	d(3) at 9%	0.7722	0.772183	-1.7e-05	match
14	4	B	102.53	102.5313	+1.3e-03	match
14	4	D	2.7390	2.738954	-4.6e-05	match
14	4	Dadj	2.5128	2.512801	+1.4e-06	match
14	4	actualB	100.000	100.00000	-2.8e-14	match
15	4	bonds price	101.464	101.46376	-2.4e-04	match
15	4	bond duration	2.73753	2.7375287	-1.3e-06	match
15	4	bond convexity	7.86779	7.8677927	+2.7e-06	match
15	4	new bond price	104.282	104.28167	-3.3e-04	match
16	5	discount factor from 1 period spot rate	0.951229	0.95122942	+4.2e-07	match
16	5	spot rate from 2 period discount factor	0.0526803	0.052680258	-4.2e-08	match
16	5	forward rate, from discount factors	0.0553605	0.055360516	+1.6e-08	match
16	5	forward rate, from yields	0.0553605	0.055360516	+1.6e-08	match
17	5	flat: discount factor t=1	0.951229	0.95122942	+4.2e-07	match
17	5	flat: discount factor t=2	0.904837	0.90483742	+4.2e-07	match
17	5	flat: spot rate t=1	0.05	0.0500	+0.0e+00	match
17	5	flat: spot rate t=2	0.05	0.0500	+0.0e+00	match
17	5	flat: forward rate 1 to 2	0.05	0.0500	+6.9e-17	match
18	5	d(1)	0.9600	0.960000	+0.0e+00	match
18	5	d(2)	0.9400	0.940000	+0.0e+00	match
18	5	d(3)	0.9200	0.920000	+0.0e+00	match

Box	Ch.	Quantity	Book	Port	Difference / note	Result
18	5	d(4)	0.8164	0.816364	-3.6e-05	match
18	5	d(5)	0.7399	0.739899	-1.0e-06	match
18	5	d(6)	0.6662	0.666180	-2.0e-05	match
18	5	r(1) (inline arithmetic)	0.085069	0.08506944	+4.4e-07	match
18	5	r(2) (inline arithmetic)	0.063830	0.06382979	-2.1e-07	match
18	5	r(3) (inline arithmetic)	0.057162	0.05716177	-2.3e-07	match
18	5	r(4) (inline arithmetic)	0.106772	0.10677203	+2.7e-08	match
18	5	r(5) (inline arithmetic)	0.105628	0.10562841	+4.1e-07	match
18	5	r(6) (inline arithmetic)	0.106884	0.10688392	-8.1e-08	match
19	5	interpolated yield t=0.1	0.1	0.100	+0.0e+00	match
19	5	interpolated yield t=0.5	0.2	0.200	+0.0e+00	match
19	5	interpolated yield t=1	0.3	0.300	+0.0e+00	match
19	5	interpolated yield t=3	0.35	0.3500	+0.0e+00	match
19	5	interpolated yield t=5	0.4	0.400	+0.0e+00	match
19	5	interpolated yield t=10	0.5	0.500	+0.0e+00	match
20	5	interpolated: discount factor t=1	0.932394	0.93239382	-1.8e-07	match
20	5	interpolated: discount factor t=2	0.865022	0.86502229	+2.9e-07	match
20	5	interpolated: spot rate t=1	0.07	0.0700	+0.0e+00	match
20	5	interpolated: spot rate t=2	0.0725	0.072500	+1.4e-17	match
20	5	interpolated: forward rate 1 to 2	0.075	0.07500	-1.4e-17	match
21	5	price	99.1088	99.108757	-4.3e-05	match
21	5	duration	1.9087	1.908703	+2.6e-06	match
21	5	convexity	3.72611	3.7261077	-2.3e-06	match
22	6	er	0.096667	0.09666667	-3.3e-07	match
22	6	sigma	0.2236	0.223607	+6.8e-06	match
23	6	a	2.1333	2.133333	+3.3e-05	match
23	6	b	0.2137	0.213667	-3.3e-05	match
23	6	c	21.667	21.66667	-3.3e-04	match
23	6	d	0.078333	0.07833333	+3.3e-07	match
23	6	g(1)	0.021277	0.02127660	-4.0e-07	match
23	6	g(2)	-2.680851	-2.68085106	-6.4e-08	match
23	6	g(3)	3.659574	3.65957447	+4.7e-07	match

Box	Ch.	Quantity	Book	Port	Difference / note	Result
23	6	h(1)	2.1277	2.127660	-4.0e-05	match
23	6	h(2)	31.9149	31.914894	-6.4e-06	match
23	6	h(3)	-34.0426	-34.042553	+4.7e-05	match
23	6	w(1)	0.2128	0.212766	-3.4e-05	match
23	6	w(2)	0.1915	0.191489	-1.1e-05	match
23	6	w(3)	0.5957	0.595745	+4.5e-05	match
24	6	mean	0.075	0.07500	+1.4e-17	match
24	6	variance	0.5	0.500	+0.0e+00	match
24	6	stdev	0.707107	0.70710678	-2.2e-07	match
25	6	w1	0.5	0.500	+2.2e-16	match
25	6	w2	0.5	0.500	-3.3e-16	match
26	7	futures price	105.127	105.12711	+1.1e-04	match
27	8	one period european call	3.64342	3.6434189	-1.1e-06	match
27	8	two period european call	5.44255	5.4425456	-4.4e-06	match
28	9	c	10.308	10.30815	+1.5e-04	match
29	9	Black Scholes call price	5.45325	5.4532452	-4.8e-06	match
30	9	Delta	0.633737	0.63373728	+2.8e-07	match
30	9	Gamma	0.0354789	0.035478872	-2.8e-08	match
30	9	Theta	-6.61473	-6.6147349	-4.9e-06	match
30	9	Vega	13.3046	13.304577	-2.3e-05	match
30	9	Rho	13.1168	13.116809	+9.3e-06	match
31	9	implied volatility, Newton	0.0500427	0.050042655	-4.5e-08	match
31	9	implied volatility, bisections	0.0500419	0.050041902	+1.6e-09	match
32	10	warrant price	10.142	10.14198	-2.0e-05	match
33	11	european call, continuous dividend	11.7344	11.734353	-4.7e-05	match
33	11	european call, discrete dividend	11.8094	11.809394	-5.6e-06	match
34	11	american call with one dividend	10.0166	10.016626	+2.6e-05	match
35	11	european futures option (book label: call; program prices the put)	0.851476	0.85147603	+3.2e-08	match
36	11	european currency call option	2.22556	2.2255642	+4.2e-06	match
37	11	perpetual call price	19.4767	19.476749	+4.9e-05	match
38	12	european call	14.9505	14.950510	+9.7e-06	match
38	12	american call	14.9505	14.950510	+9.7e-06	match

Box	Ch.	Quantity	Book	Port	Difference / note	Result
38	12	american put	6.54691	6.5469119	+1.9e-06	match
39	12	C	14.951	14.95051	-4.9e-04	match
39	12	P	6.5469	6.546912	+1.2e-05	match
40	12	delta	0.699792	0.69979217	+1.7e-07	match
40	12	gamma	0.0140407	0.014040695	-4.8e-09	match
40	12	theta	-9.89067	-9.8906719	-1.9e-06	match
40	12	vega	34.8536	34.853623	+2.3e-05	match
40	12	rho	56.9652	56.965239	+3.9e-05	match
41	12	call, continuous dividend payout	13.5926	13.592609	+9.2e-06	match
41	12	call, proportional dividend yields	11.8604	11.860394	-5.7e-06	match
41	12	call, discrete dividend amounts	12.0233	12.023254	-4.6e-05	match
42	12	futures call option, binomial	5.74254	5.7425385	-1.5e-06	match
43	12	currency call option, binomial	2.23129	2.2312871	-2.9e-06	match
44	13	explicit finite differences, european put	4.03667	4.0366684	-1.6e-06	match
44	13	explicit finite differences, american put	4.25085	4.2508507	+6.5e-07	match
45	13	black scholes put price	4.35166	4.3516596	-4.2e-07	match
45	13	implicit Euro put price	4.34731	4.3473102	+2.0e-07	match
45	13	implicit American put price	4.60064	4.6006395	-4.7e-07	match
46	13	P	4.6006	4.600640	+4.0e-05	match
47	14	call: black scholes price	14.9758	14.975798	-1.6e-06	match
47	14	call: simulated price	14.8404	14.840390	-1.0e-05	match
47	14	put: black scholes price	5.45954	5.4595402	+2.5e-07	match
47	14	put: simulated price	5.74588	5.7458758	-4.2e-06	match
48	14	call: bs delta	0.700208	0.70020841	+4.1e-07	match
48	14	call: sim delta	0.701484	0.70148362	-3.8e-07	match
48	14	put: bs delta	-0.299792	-0.29979159	+4.1e-07	match
48	14	put: sim delta	-0.307211	-0.30721122	-2.2e-07	match
49	14	Black Scholes call option price	14.9758	14.975798	-1.6e-06	match
49	14	Simulated call option price	14.995	14.99499	-1.2e-05	match
49	14	Black Scholes put option price	5.45954	5.4595402	+2.5e-07	match
49	14	Simulated put option price	5.5599	5.559899	-9.7e-07	match
50	14	Black Scholes call option price	14.9758	14.975798	-1.6e-06	match

Box	Ch.	Quantity	Book	Port	Difference / note	Result
50	14	Simulated call option price	14.995	14.99499	-1.2e-05	match
50	14	Simulated call option price, CV	14.9758	14.975798	-1.6e-06	match
50	14	Simulated call option price, AV	14.9919	14.991933	+3.3e-05	match
50	14	Black Scholes put option price	5.45954	5.4595402	+2.5e-07	match
50	14	Simulated put option price	5.41861	5.4186106	+5.7e-07	match
50	14	Simulated put option price, CV	5.42541	5.4254113	+1.3e-06	match
50	14	Simulated put option price, AV	5.46043	5.4604289	-1.1e-06	match
51	14	cash or nothing, Q=1	0.547427	0.54742664	-3.6e-07	match
51	14	cash or nothing, control variate	1.02552	1.0255155	-4.5e-06	match
51	14	cash or nothing, antithetic variate	0.549598	0.54959825	+2.5e-07	match
51	14	asset or nothing	70.5292	70.529233	+3.3e-05	match
51	14	asset or nothing, control variate	69.8451	69.845139	+3.9e-05	match
51	14	asset or nothing, antithetic variate	70.2205	70.220536	+3.6e-05	match
52	15	American put, Johnson approximation	0.11582	0.1158196	-4.2e-07	match
53	15	Call price, Barone-Adesi Whaley (listing as printed)	5.74339	5.7433853	-4.7e-06	match
54	16	Bermudan put price	15.8869	15.886942	+4.2e-05	match
55	16	Analytical geometric average	6.74876	6.7487615	+1.5e-06	match
56	16	Lookback call price	27.0713	27.071344	+4.4e-05	match
57	16	simulated arithmetic average	1.49696	1.4969564	-3.6e-06	match
57	16	simulated geometric average	1.38017	1.3801670	-3.0e-06	match
57	16	analytical lookback put	16.2665	16.266460	-4.0e-05	match
57	16	simulated lookback put	14.9846	14.984649	+4.9e-05	match
57	16	analytical lookback call	22.8089	22.808887	-1.3e-05	match
57	16	simulated lookback call	21.9336	21.933650	+5.0e-05	match
57	16	simulated lookback call, control variates	22.0685	22.068467	-3.3e-05	match
58	17	american call price	14.9505	14.950510	+9.7e-06	match
58	17	american put price	6.54691	6.5469119	+1.9e-06	match
59	17	binary option price	0.498858	0.49885766	-3.4e-07	match
60	18	american put, trinomial	6.52719	6.5271874	-2.6e-06	match
61	19	Merton jump diffusion call	23.2074	23.207433	+3.3e-05	match
62	19	heston call price	3.06944	3.0694370	-3.0e-06	match
63	20	zero coupon put option price	1.92791	1.9279054	-4.6e-06	match

Box	Ch.	Quantity	Book	Port	Difference / note	Result
63	20	coupon bond put option price	2.22852	2.2285235	+3.5e-06	match
63	20	zero coupon american put, binomial	2.43282	2.4328243	+4.3e-06	match
64	21	Debt value	81.8592	81.859230	+3.0e-05	match
65	22	Nelson Siegel: direct calculation, yield	0.0363142	0.036314232	+3.2e-08	match
65	22	Nelson Siegel: yield (t=1)	0.0363142	0.036314232	+3.2e-08	match
65	22	Nelson Siegel: discount factor (t=1)	0.964337	0.96433722	+2.2e-07	match
65	22	Nelson Siegel: forward rate (1, 2)	0.0300602	0.030060164	-3.6e-08	match
66	22	cubic spline: direct, discount factor (t=1)	1.1	1.100	+0.0e+00	match
66	22	cubic spline: yield (t=1)	-0.0953102	-0.095310180	+2.0e-08	match
66	22	cubic spline: discount factor (t=1)	1.1	1.100	+0.0e+00	match
66	22	cubic spline: forward (1, 2)	0.318454	0.31845373	-2.7e-07	match
67	22	CIR: direct, discount factor (t=1)	0.951166	0.95116586	-1.4e-07	match
67	22	CIR: yield (t=1)	0.0500668	0.050066830	+3.0e-08	match
67	22	CIR: discount factor (t=1)	0.951166	0.95116586	-1.4e-07	match
67	22	CIR: forward (1, 2)	0.0498756	0.049875633	+3.3e-08	match
68	22	Vasicek: direct, discount factor (t=1)	0.955408	0.95540798	-2.1e-08	match
68	22	Vasicek: yield (t=1)	0.0456168	0.045616826	+2.6e-08	match
68	22	Vasicek: discount factor (t=1)	0.955408	0.95540798	-2.1e-08	match
68	22	Vasicek: forward rate (1, 2)	0.0281476	0.028147632	+3.2e-08	match
69	23	Rendleman Bartter option on zero coupon bond	0.00713661	0.0071366126	+2.6e-09	match
70	24	tree time 0	0.1	0.100	+0.0e+00	match
70	24	tree time 1, node 0	0.099	0.09900	+0.0e+00	match
70	24	tree time 1, node 1	0.102	0.10200	+1.4e-17	match
70	24	tree time 2, node 0	0.09801	0.0980100	+0.0e+00	match
70	24	tree time 2, node 1	0.10098	0.1009800	+0.0e+00	match
70	24	tree time 2, node 2	0.10404	0.1040400	+1.4e-17	match
71	24	Bond price B	98.5997	98.599716	+1.6e-05	match
72	24	Straight bond price	89.9017	89.901746	+4.6e-05	match
72	24	Callable bond price	89.2483	89.248266	-3.4e-05	match
73	25	Ho-Lee call, flat term structure	6.4857	7.098447	+6.1e-01	DIFFERS
73	25	Ho-Lee call, Nelson Siegel term structure	6.35411	6.9528190	+6.0e-01	DIFFERS
74	26	Vasicek call option price	0.000226833	0.00022683258	-4.2e-10	match

Box	Ch.	Quantity	Book	Port	Difference / note	Result
75	27	Boost date and time demo (4 printed lines)	date: 2014-Apr-01 ...	n/a	Demonstrates the Boost C++ library and prints the current clock time; not ported	not comparable
76	A	N(0)	0.5	0.500	-2.4e-08	match
76	A	N(0,0,0)	0.25	0.2500	-1.4e-09	match
77	A	random uniform number 1	0.840188	0.84018772	-2.8e-07	match
77	A	random uniform number 2	0.394383	0.39438293	-7.3e-08	match
77	A	random uniform number 3	0.783099	0.78309922	+2.2e-07	match
77	A	random uniform number 4	0.79844	0.7984400	+3.3e-08	match
77	A	random uniform number 5	0.911647	0.91164736	+3.6e-07	match
77	A	random normal number 1	-1.07224	-1.0722428	-2.8e-06	match
77	A	random normal number 2	0.925946	0.92594602	+2.2e-08	match
77	A	random normal number 3	2.70202	2.7020241	+4.1e-06	match
77	A	random normal number 4	1.36918	1.3691849	+4.9e-06	match
77	A	random normal number 5	0.0187313	0.018731341	+4.1e-08	match